

System Testing and Validation Cases

The following test suites were executed to validate system stability and functionality.

Functional Test Results

TC-01: Website Screen Test

- **What we tested:** Opening the home page.
- **What we did:** Opened the main website address.
- **What should happen:** The page opens with all design elements (canvas, background graphics, side menus).
- **Result:**  Pass

TC-02: Backend Connection Test

- **What we tested:** Getting data from the system.
- **What we did:** Sent a request to `/api/research`.
- **What should happen:** The system replies with structured information (answers, key concepts, next steps).
- **Result:**  Pass

TC-03: Settings & Mode Test

- **What we tested:** Checking if special settings work.
- **What we did:** Selected the "Synthesis" mode.
- **What should happen:** The system uses the correct instructions for synthesis mode.
- **Result:**  Pass

- **TC-04: Error Warning Test**

- **What we tested:** Checking how the system handles mistakes.
- **What we did:** Submitted a blank/empty search.
- **What should happen:** The system shows an error message (Error 400: Bad Request) instead of crashing.
- **Result:**  Pass

2. API Stress Verification

Running concurrent queries to check socket limits.

- Maximum capacity is rate-limited by standard API endpoints (~15 requests per minute).
- Invalid configurations yield correct HTTP error messages.